

GHAYDA NA'IM JA'AFREH

AI / Backend Engineer | Python | FastAPI | REST APIs | Model Integration

Karak, Jordan | ghayda.jaafreh@gmail.com
linkedin.com/in/ghayda-njaafreh | github.com/ghayda-njaafreh | kaggle.com/jaafreh

PROFESSIONAL SUMMARY

Data Science and Artificial Intelligence graduate with an Excellent GPA of 93.04% and hands-on experience building, integrating, debugging, and documenting Python-based AI applications and backend workflows. Strong foundation in FastAPI, REST APIs, authentication logic, modular programming, validation, error handling, logging, dependency management, and model-service integration. Built applied systems involving multimodal authentication, healthcare AI, computer vision, robotics integration, and intelligent game applications. Microsoft Certified in Azure AI Fundamentals with practical exposure to Azure-hosted GPU infrastructure.

TECHNICAL SKILLS

Backend & Software Engineering: Python, FastAPI, REST APIs, API integration, modular programming, input validation, error handling, logging, debugging, dependency management, virtual environments

AI & Model Integration: PyTorch, TensorFlow, Keras, scikit-learn, model inference, classification, CNNs, segmentation, explainable AI, model evaluation

Data & Databases: SQL, SQLite, MySQL, Pandas, NumPy, structured data workflows

Security & Authentication: Voice verification, multimodal authentication workflows, privacy-aware AI, secure data handling, networking and cybersecurity fundamentals

Engineering Practices: Interface validation, incremental testing, reproducible experimentation, technical documentation, architecture/workflow diagrams, Git, GitHub

Tools & Platforms: OpenCV, SpeechBrain, InsightFace, Vosk, Gradio, Jupyter Notebook, Azure-hosted GPU environment, Pygame

EDUCATION

Bachelor's Degree in Data Science and Artificial Intelligence | Mutah University | 2022-2026

GPA: 93.04% | Grade: Excellent | Ranked 2nd in the Data Science and Artificial Intelligence graduating cohort

SELECTED PROJECTS

Privacy-Safe Multimodal Authentication System

Team Project | Main Voice-Model and Integration Contributor

- Led work on the voice-verification component and contributed to integration, verification logic, API-related workflow, testing, logging, and privacy-aware data handling.
- Integrated voice verification with a Python/FastAPI multimodal workflow that also incorporated face verification and optional offline speech recognition; resolved runtime, dependency, and input-format issues through systematic testing.

Technologies: Python, FastAPI, PyTorch, SpeechBrain, OpenCV, InsightFace, Vosk, REST APIs, authentication, logging

StrokeLens - Secure & Explainable Healthcare AI Graduation Project

Graduation Project | Sep 2025-Jun 2026 | Core Project Owner and Lead Technical Contributor

- Led an end-to-end healthcare AI project using structured clinical data, multi-stage stroke-risk modeling, explainability, secure reporting, and integrated application workflows.
- Contributed across data preparation, model experimentation and evaluation, threshold analysis, explainability, system integration, testing, technical documentation, and Azure-hosted GPU experimentation.

Technologies: Python, machine learning, explainable AI, model evaluation, secure systems, Azure GPU

Tetris AI Lab - End-to-End AI Game Application

Individual Project | Public GitHub Release

- Developed and published a Python/Pygame application integrating a CNN policy with inference, top-5 one-piece lookahead, safety controls, replay functionality, and human/AI gameplay modes.
- Built a reproducible training-data pipeline with automated cleaning, video-level train/validation/test splitting, simulator-generated recovery states, simulator-verified mirror augmentation, experiment tracking, testing, and release documentation.

Technologies: Python, PyTorch, CNN, Pygame, data pipelines, testing, reproducible experimentation

Dr. NAO - AI-Assisted Dermatology Support System

Team Project | Main AI/Model and Robot-Integration Contributor

- Integrated AI model outputs with a NAO robot interaction workflow for a healthcare prototype presented in a live academic demonstration.
- Supported model and robot integration, interaction logic, testing, troubleshooting, demonstration preparation, and technical explanation.

Technologies: Python, computer vision, AI integration, NAO robotics, testing, troubleshooting

Medical Image Segmentation & Computer Vision Portfolio

Kaggle and Academic Projects

- Built teeth instance segmentation for dental X-rays, achieving Mask mAP50 of 0.9947 and Mask mAP50-95 of 0.6301.
- Implemented a custom PyTorch U-Net workflow for dental-caries segmentation, achieving 0.9010 test Dice; additional work includes Intel transfer learning, MNIST CNN experiments, OCR with HOG/KNN, and image classification.

Technologies: Python, PyTorch, OpenCV, U-Net, transfer learning, OCR, segmentation

EXPERIENCE

Data Science & Programming Trainee | The Hope International Company | Online Training | Jun 20, 2025-Sep 10, 2025

- Completed structured practical training in Python programming, data analysis, preprocessing, visualization, model implementation, debugging, teamwork, and technical documentation.
- Applied analytical and programming skills to structured datasets and small end-to-end AI workflows.

AI & Machine Learning Trainee | The Hope International Company | Online Training | Nov 15, 2023-Mar 15, 2024

- Built and evaluated Python-based machine-learning exercises covering data preparation, feature selection, classification, regression, decision trees, KNN, Naive Bayes, and model evaluation.
- Developed end-to-end understanding of the machine-learning lifecycle from preparation through evaluation and reporting.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Microsoft Certified: Azure AI Fundamentals (AI-900) | Microsoft | Earned May 27, 2026 | Credential ID: C5834AFC20D9A41C | Certification No.: B010AT-7276E5
- Data Science Essentials with Python | Cisco Networking Academy | May 29, 2026 | Certificate ID: 960f2b70-a252-42a1-a73d-13e7ca5c7c51
- Python Essentials 1 | Cisco Networking Academy | May 30, 2026 | Certificate ID: 01720f08-f6be-48a8-b730-d4da54339d1f
- Deep Learning - 60 Hours | The Hope International Company | Sep 15, 2025 | Certificate ID: 1758145099980
- Python AI (Machine Learning) - 60 Hours | The Hope International Company | Nov 2023-Mar 2024
- Ethical Hacker | Cisco Networking Academy | May 29, 2026 | Certificate ID: 34898e2b-05cf-41a1-bcc0-c8d3f49090d1
- Cyber Security Associate - Certificate of Completion | Green Circle for Software Solutions | 2026
- CCNA Course Certificate | Pioneers Academy | Dec 28, 2023 | Certificate No.: 18437657-24
- Security+ Course Certificate - 30 Hours | Pioneers Academy | Oct 5, 2023 | Certificate No.: 18424325-2560
- Human Research: Data or Specimens Only Research - Basic Course | CITI Program, MIT Affiliates | Jul 13, 2025-Jul 13, 2028 | Record ID: 70859362
- HTML Essentials | Cisco Networking Academy | May 30, 2026 | Certificate ID: 1ca22c8d-bbae-4328-8cf8-07eacac17e5e
- International Computer Driving License (ICDL) - 36 Hours | Arab International Academy | Feb 5, 2024 | CRT.No: 201548
- English Communication Skills Training - 20 Hours | Mutah University / King Abdullah II Fund for Development | Apr 6-May 13, 2024
- English Conversation - 12 Levels, 90 Hours | Arab International Academy | Feb 1-May 15, 2023 | ID: 200166016 | RN: 117840

LANGUAGES & CAREER FOCUS

Languages: Arabic (Native), English (Intermediate)

Career Focus: AI Backend Engineering, Python Backend Development, REST APIs, Model-Service Integration, Authentication Workflows, Applied AI Systems, and AI Product Development.